



Trading Credit Events in Corporate Bond Markets

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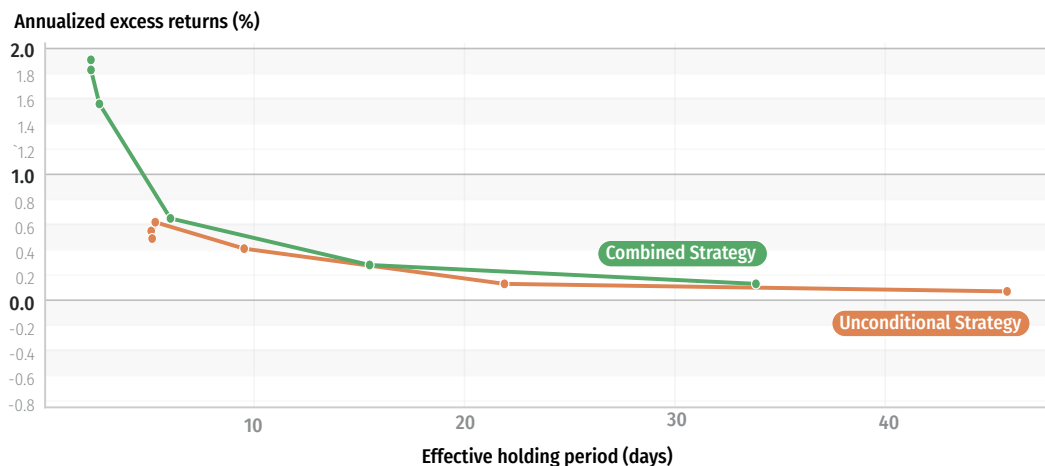
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Executive Summary

In this paper, we evaluate the impact of credit-related announcements on corporate bond prices, conditioning news on past bond performance and news sentiment. We find:

- (i) **A “surprise” effect:** Conditioning credit ratings, equity actions, and stock prices on good news and poor past performance generates significant positive abnormal bond returns. Conditioning credit ratings, analyst ratings, and price targets on bad news and good past performance generates long-lasting outperformance.
- (ii) **A “buy the rumor, sell the news” effect:** Conditioning credit ratings, equity actions, analyst ratings, price targets, and stock prices on good news and good past performance generates negative abnormal bond returns.
- (iii) **A combined strategy outperforms:** A strategy excluding good announcements associated with good past performance while targeting unexpected events outperforms the unconditional strategy and individual strategies. The combined strategy generates annualized returns of 200 bps above the annual credit benchmark at a two-day effective holding period, and 90 bps at the one-week holding period, compared to 50 bps for the unconditional strategy.



Annualized returns of the long-only combined strategy (green) in excess of the annual U.S. investment-grade credit benchmark over various effective holding periods when tilting with credit sentiment aggregated at the bond issuer level across all news. Source: RavenPack, September 2022

1. Introduction

2022 has been a challenging year for fixed income investors. With inflation at multi-decade highs globally, interest rates have surged sending bond prices sharply lower. Year to date, U.S. investment-grade corporate bonds¹ are down 17%. Looking ahead, the winding down of central banks' asset purchasing programs, and weaker economic growth, could create further headwinds for the asset class. Against this backdrop, bond selection skills are crucial in navigating rate volatility and allocating capital to the right securities.

Over the last decade, the emergence of real-time, machine-readable news has paved the way for sentiment-based trading strategies that predict investment performance by analyzing media content. The seminal work of TETLOCK (2007) and TETLOCK ET AL. (2008) involved examining the effects of news sentiment from articles in *The Wall Street Journal* on stock returns. In-sample, a one standard deviation increase in pessimism was associated with an eight basis point (bps) fall in the Dow Jones Industrial Average the following day. The research also showed that the relative occurrence of negative words in news stories is a statistically significant predictor of individual firms' investment performance (see LOUGHRAN AND MCDONALD, 2016 for a review). More recently, a study by HAFEZ ET AL. (2018) showed that signals based on RavenPack news sentiment are particularly predictive over short-term horizons, especially in the more efficient, liquid U.S. markets.

In this paper, we demonstrate that credit-related news has a significant impact on bond prices. Conditioning announcements associated with credit ratings, analyst ratings, and price targets on bad news and good past performance generates long-lasting outperformance, with news incorporated slowly into prices. Good news following poor past bond performance exhibits substantial abnormal returns on the day of the news, while also generating post-event outperformance, particularly for events associated with credit and insider trading. This suggests that, for these events, unexpected announcements generate stronger price reactions in the short term. While we observe that credit risk agencies and analysts tend to downgrade bond issuers following bad news and negative past bond performance, we find that bond markets tend to trade against positive news, demonstrating the “buy the rumor, sell the news” phenomenon.

The remainder of the paper is organized as follows: **SECTIONS 2 and 3** provide an exploratory analysis of sentiment indicators derived from credit-related news, and of our credit data sample. **SECTION 4** describes the methodology used to evaluate the market impact of announcements on bond returns at the event category level. **SECTION 5** describes how we construct and tilt the credit benchmark towards credit sentiment scores at the category level, and form long-only credit strategies while disentangling and recombining the individual effects of past bond performance and news sentiment. **SECTION 7** summarizes our findings.

1 Source: **S&P 500 Investment-Grade Corporate Bond Index**.

2. Corporate Credit Data

Our study relies on transaction records from IHS Markit iBoxx, a leading fixed income data provider, for the sample period of February 2015 to September 2022.

1. Selection Criteria

To ensure a liquid, efficient corporate bond universe, eligibility for inclusion is based on the following selection criteria as documented in BAI, BALI, AND WEN (2019), BATRAM ET AL. (2019), and BESSEMBINDER ET AL. (2009) among others:

1. The sample is limited to **U.S. dollar-denominated, senior, unsecured, investment-grade** rated corporate bonds issued by companies located in the U.S., and listed, cleared, and traded in the U.S. public market. It excludes bonds issued through private placement or under the 144A exemption rule of the U.S. Securities Act.
2. Bonds are required to be issued by **non-financial** corporate entities, as U.S. non-financial companies dominate the primary corporate bond markets, making up 35% of global issuance in 2018 compared to 20% for European companies (Çelik, 2019).
3. Bonds are required to have **fixed coupons**. We exclude bonds with **embedded options** or sinking funds, amortizing bonds, bonds with conversion options, collateralized debt obligations, floating coupon rate or step-up coupons, callable, convertible, zero-coupon, equity linked, and mortgage- or asset-backed bonds.
4. Corporate bonds are required to be **liquid** to avoid mispricings, and have a minimal amount outstanding of **USD \$500 million** for investment-grade securities, quoted prices of **between USD \$5 and USD \$1,000**, remaining time to maturity of at least **one year**, and non-negative bid-ask spreads. We eliminate bonds with **abnormal returns**, removing transaction records for securities with daily returns in excess of 20%.

We collect bond specific data including sector, yield to maturity, spread duration, option-adjusted spread (OAS), coupons, bid-ask prices, total returns, accrued interest, and asset swap spreads. **U.S. dollar-denominated total returns** for bond i in currency j are calculated as follows:

$$R_{i,t}^j = \frac{P_{i,t}^j + AI_{i,t}^j + C_{i,t-1}^j}{P_{i,t-1}^j + AI_{i,t-1}^j},$$

where $R_{i,t}^j$ is bond i 's return at time t , $P_{i,t}^j$ is the transaction price per unit of face value, $AI_{i,t}^j$ is accrued interest, and $C_{i,t}^j$ is the coupon paid between $t-1$ and t .

2. Data Description

In our sample of 1,938 trading days between February 2015 and September 2022, we observe 2,375 unique investment-grade corporate bonds from 571 distinct bond issuers, with data for approximately 983 bonds on average on any given trading day.

FIGURE 1 provides a summary of the investment-grade universe. For the AAA, AA, A, and BBB ratings, there are 38, 217, 996, and 1,124 unique corporate bonds respectively. The latter two ratings constitute 90% of U.S. investment-grade issuance, on average.

As observed by the OECD (ÇELİK, 2019), corporate bond issuance by U.S. non-financial companies has declined noticeably since 2017 as a result of the **Tax Cuts and Jobs Act**, which has lowered the corporate tax rate, and unlocked the overseas cash positions of U.S. companies by reducing the cost of repatriating foreign earnings. **FIGURE 2** (bottom) shows that since 2017, the number of U.S. non-financial companies that have raised capital via bond issuance has decreased considerably. Consequently, there has been an **increase in issuer concentration** within the U.S. primary corporate bond markets.

3. Benchmark Construction

In the remainder of the study, we construct a long-only **U.S. investment-grade credit benchmark** weighted by the market values $V_{j,t}$ associated with eligible corporate bonds. Benchmark weights are calculated as follows:

$$W_{j,t}^L = \delta_L \frac{V_{j,t}}{\sum_{i=1}^N V_{i,t}}$$

where $\delta_L = 1$ to ensure gross portfolio exposure of 100%. We apply a maximum allocation constraint of 2.5% per bond at each daily rebalancing, reallocating capital until the maximum gross exposure of 100% is achieved.

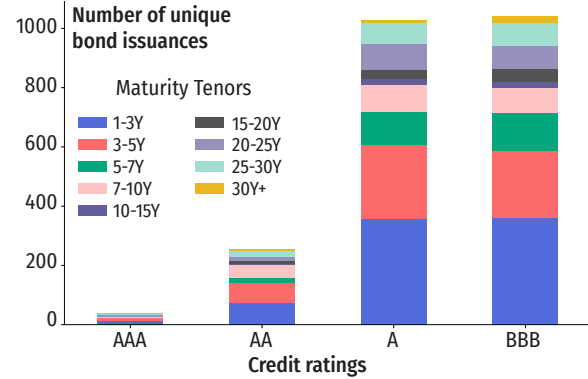


FIGURE 1: Number of bond issuances segmented by credit ratings and maturity tenors for the investment-grade universe.
Source: RavenPack, September 2022

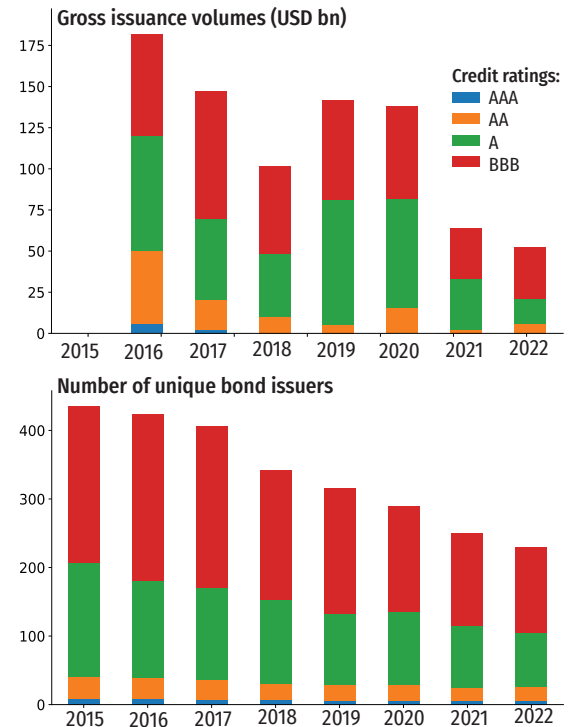


FIGURE 2: Total amount issued (top) and number of unique bond issuers (bottom) segmented by credit ratings and issuance years for the investment-grade universe.
Source: RavenPack, September 2022

3. Credit Events

We select credit-related events at a granular level based on the following RavenPack Edge filters:

- **Credit Sentiment Score:** *Credit Sentiment Score* $\neq 0$ to ensure we select non-neutral events.
- **Groups of Events:** Credit events that have a well documented impact on bond issuers, in terms of:
 - Cash flow generation: *Earnings, Revenues, Dividends, Assets*
 - Credit quality: *Credit, Bankruptcies, Credit Ratings, Analyst Ratings*
 - Credit sentiment: *Insider Trading, Stock Prices, Price Targets, Stock Picks, Technical Analysis*
 - Capital structure: *Equity Actions, Acquisitions & Mergers*
- **Source Providers:** News, Transcripts, EDGAR filings.
- **Event Relevance:** Event is mentioned in the headline (*Event relevance* ≥ 90) for all document types except for Transcripts and EDGAR filings.
- **Entity Relevance:** Entity is mentioned in the headline (*Entity relevance* ≥ 90) for Transcripts and EDGAR filings only, as these contain no relevant events in the headline, aside from the company name that is discussed in the text body.
- **Global Event Similarity Days:** We focus on novel events and avoid repeated events. We aggregate by company, event similarity key, and category. We calculate the day difference between the current and most recent observation and retain novel events defined as *Event Similarity Days* ≥ 90 .
- **Source Rank:** Only high quality and reputable sources of news (*Source Rank* ≤ 4).

FIGURE 3 (top) shows that over 40% of the credit events capture yearly information related to corporate earnings. Bankruptcies represent only 236 events over the 2015-2022 period, yet are associated with strong negative sentiment.

Corporate earnings and revenues are prominent in newsflashes and transcripts, often from top-ranked sources. By contrast, mergers & acquisitions and analyst reports are generally associated with full articles, often from lower-ranked sources. Events related to insider trading and equity actions are primarily captured in SEC filings and tabular material as bond issuers are required to report these to regulators.

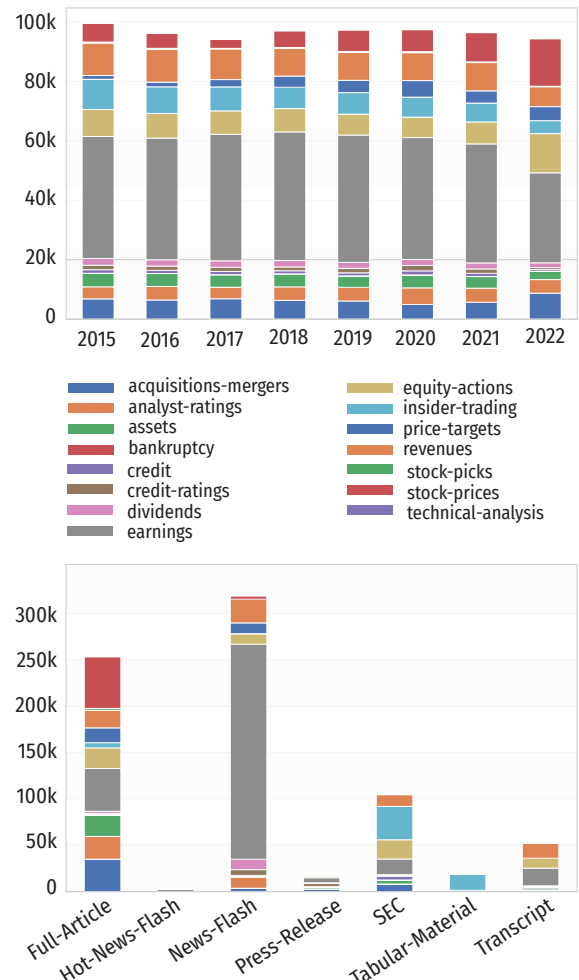


FIGURE 3: Distribution of the volume of events across time (top) segmented by document type (bottom) across groups of credit events in the RavenPack taxonomy. Source: RavenPack, September 2022

4. Credit Event Studies

We conduct an event study to evaluate the price impact of granular credit events on corporate credit markets, following BESSEMBINDER ET AL. (2009) and EDERINGTON ET AL. (2015) among others.

4.1 Weighted Abnormal Bond Returns

Credit event studies often fail to control for high cross-sectional heteroskedasticity as bond return volatility tends to increase for long-dated, low-rated, and illiquid bonds, resulting in low test power. Accordingly:

1. We calculate **abnormal bond returns** using 16 value-weighted ratings (AAA, AA, A, BBB) and maturity benchmark portfolios (1 to 3 years, 3+ to 5 years, 5+ to 10 years, and over 10 years) across the investment-grade universe.

$$ABR_j[t-1, t+1] = R_j[t-1, t+1] - BM_n[t-1, t+1]$$

where $BM_n[t-1, t+1]$ is the mean return on a benchmark portfolio n matching the rating and the maturity of bond j .

2. We aggregate bonds **at firm level**, as returns for bonds issued by the same company tend to be highly correlated, leading to biased test statistics. We combine firms' bond returns into a single firm return by weighting bond returns by the volume of bonds outstanding relative to the company total within our universe.

$$WABR_j[t-1, t+1] = \sum_{i=1}^J ABR_i[t-1, t+1] \cdot w_i$$

where J is the number of bonds outstanding for firm j and w_i is the market value weight of bond i relative to the total market value of bonds outstanding for firm j .

4.2 Cumulative Average Abnormal Bond Returns

For each group of credit events, we aggregate the credit sentiment scores at the issuer level on a daily basis, by computing the arithmetic average. This results in one signal per day for each bond issuer and events group. The sign of this signal is used to distinguish between **“good”** and **“bad”** news. This allows us to group events into two different news types. We then compute the cumulative average abnormal returns, $CAAR$, for each day of the event window. As displayed in **FIGURE 4**, we compute two separate cumulative performances around the event date to assess its market impact on bond returns.

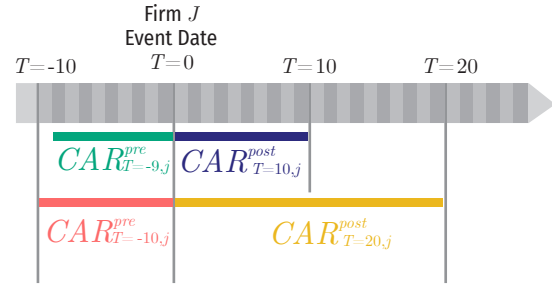


FIGURE 4: Calculation of the pre-event and post-event cumulative abnormal returns, throughout the event window.

1. First, we calculate the **pre-event performance** $CAR_{T,j}^{pre}$ for the bond issuer j , that the investor would generate from $T \in [-10, -1]$ to $T = 0$ (close-to-close returns) as:

$$CAR_{T,j} = \sum_{t=-10}^T WABR_{j,t} \quad \forall T \in [-10, 0]$$

$$CAR_{T,j}^{pre} = CAR_{T,j} - CAR_{T=0,j}$$

such that we observe $CAR_{T=0,j}^{pre}=0$, an increasing function if cumulative performance is positive (note that values will be negative) and a decreasing function otherwise.

2. Second, we calculate the **post-event performance** $CAR_{T,j}^{post}$ for the bond issuer j , that the investor would generate from $T=1$ onwards as:

$$CAR_{T,j}^{post} = \sum_{t=1}^T WABR_{j,t} \quad \forall T \in [1, 20]$$

3. Third, for each group of events, news type (“good” or “bad”) and trading day in the event window, we compute the **cumulative average abnormal return**, $CAAR_T$, as the arithmetic mean of all available for that event day:

$$CAAR_T = \frac{\sum_{i=1}^N CAR_{T,i}}{N} \quad \forall T \in [-10, 20]$$

$\wedge N$ = total observations on event day T

In this last step, we calculate the mean in the time series and cross section simultaneously, ensuring that we have a large enough sample size with at least five observations per event day.

4.3 Unconditional Event Studies

FIGURE 5 shows asymmetry between good and bad news, as bond price reaction is significantly stronger for the latter. This confirms that markets are more sensitive to bad news. Interestingly, the credit event study exhibits a significant drift in performance prior to the realization of the bad news, suggesting that investors anticipate the news. This is particularly the case for events related to **credit ratings**, price targets, analyst ratings, earnings, and credit. This is logical as credit rating downgrades and credit restructuring often follow weaker-than-expected corporate earnings.

When analyzing groups individually, events related to price targets have the most impact, with nearly 20 bps underperformance 20 days after bad news and nearly 5

bps outperformance after good news. Events associated with stock prices and analyst ratings exhibit a price reaction only with bad news, with nearly 10 bps and 8 bps underperformance after 10 and 20 days, respectively. Interestingly, bad news associated with credit ratings shows a negative price reaction of nearly 50 bps prior to the news, with a price reversal 20 days after the day of the news. Mergers & acquisitions, stock picks, and technical analysis do not have a significant impact on returns. Counterintuitively, events associated with insider trading and credit exhibit underperformance (outperformance) in the presence of good (bad) news. This requires a more granular investigation of bond price reaction at the category level within each group of events, and a filtering out of event categories that have less impact on bond prices.

- **Credit group:** KOLODNY AND SUHLER (1998) found that new debt issuance announcements have a significant positive impact on existing bondholders' returns while AKHIGBE, EASTERWOOD, AND PETIT (1997) documented a more complex relationship based on the credit quality of the bond issuer. More specifically, they demonstrated a negative bond price reaction to news of a new debt offering when bond issuers were facing an unexpected deterioration in cash flow generation. Existing bondholders tend to react negatively to higher leverage ratios when the bond issuer is in distress. **FIGURE 6** shows that announcements related to **debt extension** and **debt sale** (debt sale considered, debt sale rumor, debt sale) have the most significant price reactions, with nearly 20 bps and 25 bps (for debt sale considered) respectively 20 days following the news. Announcements of debt extensions and sales usually lead to positive reactions. Extending bond maturity allows issuers to better manage their outflows, while selling debt signals demand from investors.

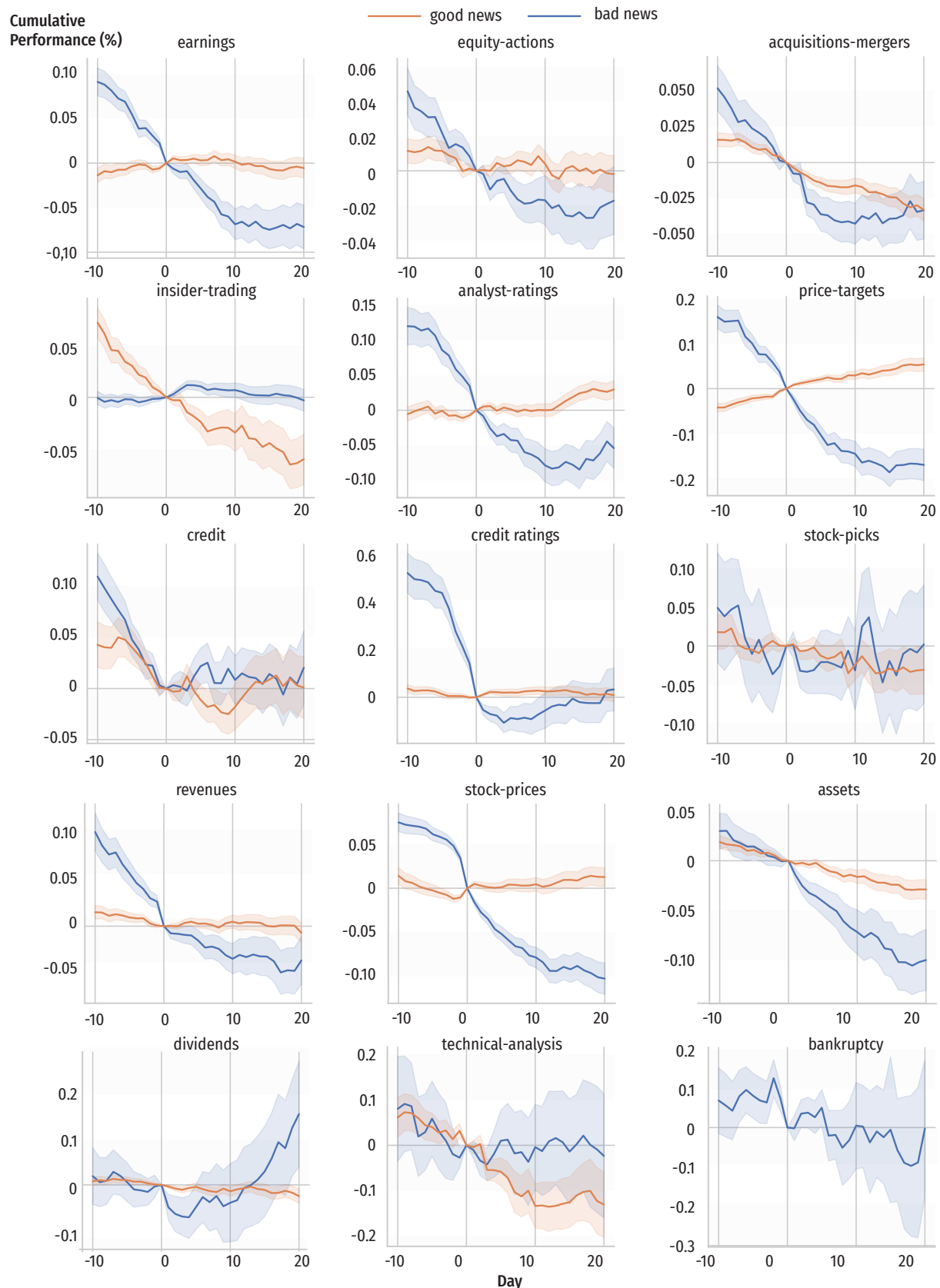


FIGURE 5: Cumulative average abnormal returns, *CAAR*, throughout the event window for good and bad news, across groups of credit-related events. Shaded areas correspond to +/- 1 standard deviation confidence intervals. Bond returns are in excess of a bond benchmark of the same maturity and credit rating. Source: RavenPack, September 2022

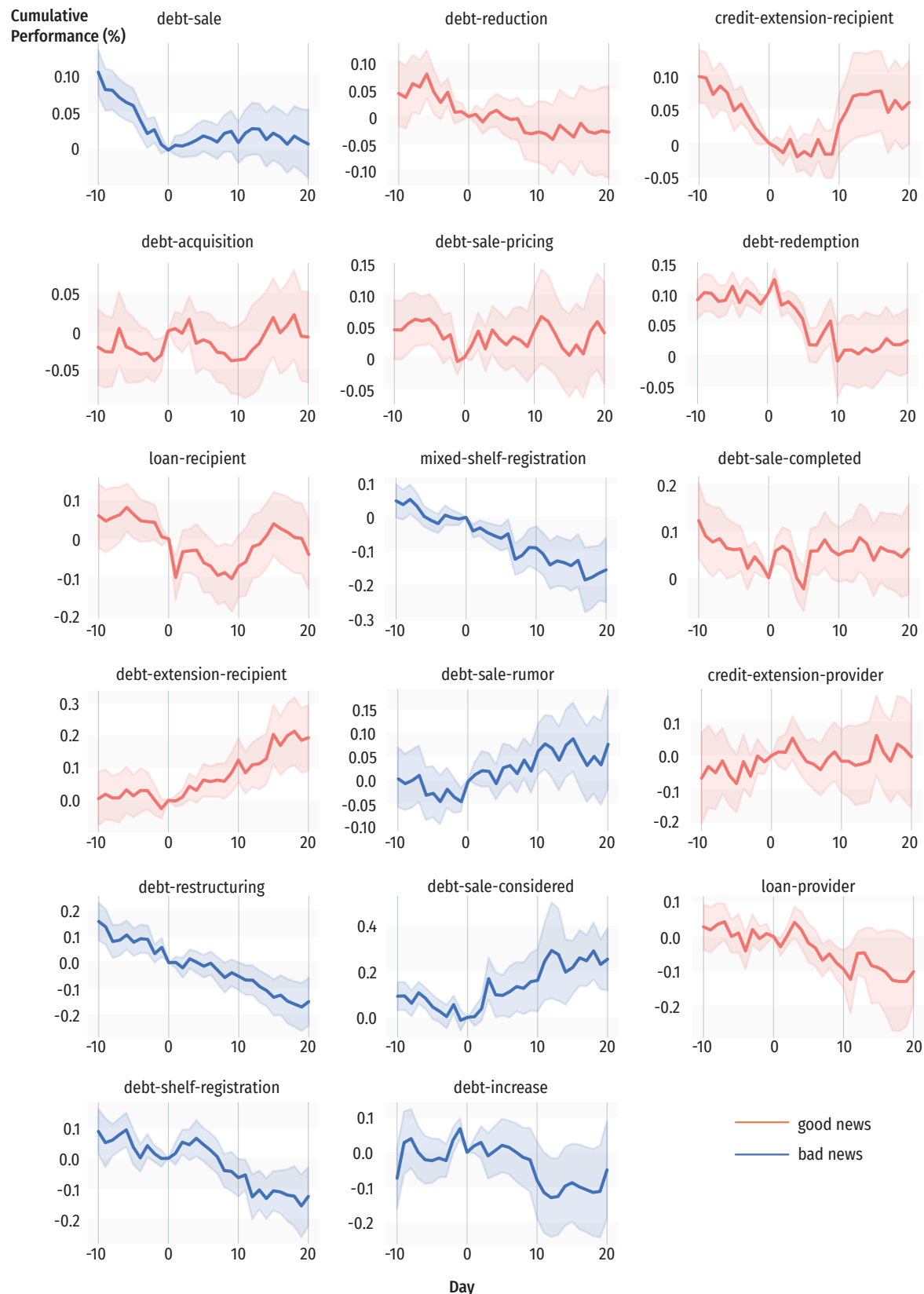


FIGURE 6: Cumulative average abnormal returns, CAAR, throughout the event window for good and bad news, across events categorized in the credit group. Shaded areas correspond to +/- 1 standard deviation confidence intervals. Bond returns are in excess of a bond benchmark of the same maturity and credit rating. Source: RavenPack, September 2022

FIGURE 7 shows that when retaining only events associated with debt extension and debt sales,¹ there is a reversal in bond performance following the announcement. This leads to pre- and post-performance of -10 bps and 5 bps versus a benchmark with the same maturity and credit rating, respectively.

- **Insider trading group:** To retain events from official reporting sources only, we filter exclusively for EDGAR files. As observed in **FIGURE 8** and in line with OEHLER ET AL. (2016), news of insider purchases negatively influences corporate bond prices whereas news of insider sales tends to have a positive impact. This counterintuitive effect reflects the fact that bondholders usually demand a higher bond premium to compensate for their disadvantage when insiders exploit private information by selling bonds.

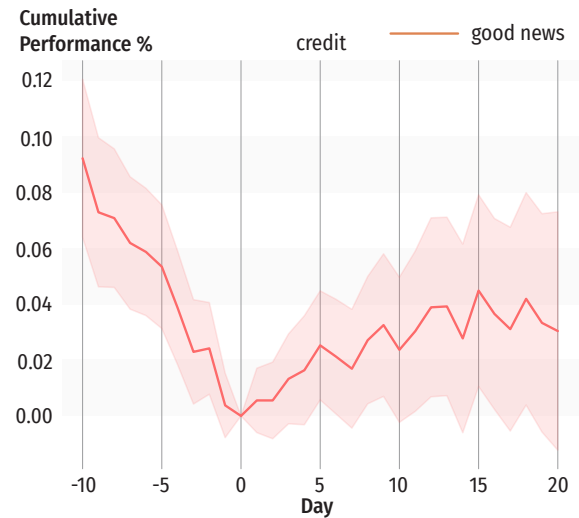
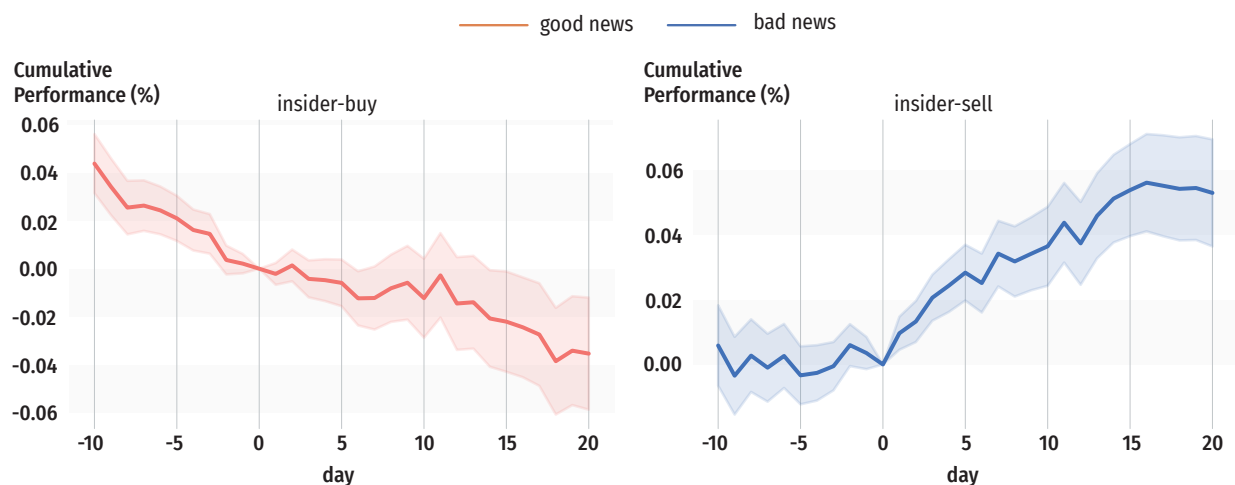


FIGURE 7: Cumulative average abnormal returns, *CAAR*, throughout the event window for good news, across events from the curated credit group. Shaded areas correspond to ± 1 standard deviation confidence intervals. Bond returns are in excess of a bond benchmark of the same maturity and credit rating. Source: RavenPack, September 2022

FIGURE 8: Cumulative average abnormal returns, *CAAR*, throughout the event window for good and bad news, across events associated with insider trading. Shaded areas correspond to ± 1 standard deviation confidence intervals. Bonds returns are in excess of a bond benchmark of the same maturity and credit rating. Source: RavenPack, September 2022



¹ We invert the sign for debt sales, in line with Kolodny and Suhler (1998), resulting in only good events associated with the curated credit category.

4.4 Conditional Event Studies

To control for other events, we condition the event studies on past bond performance, which theoretically reflects all information. For illustration, a credit downgrade usually follows weak corporate earnings, while debt issuance will be perceived as positive for low-leverage bond issuers raising capital expenditures for new investments. Bondholders tend to react more strongly following an announcement they were not expecting. This should be the case when bad news is announced following good past performance.

FIGURE 9 shows that bad news tends to generate stronger bond price reactions than good news, as observed in the conditional event studies. When following securities with poor past performance, bond markets tend to react more severely to bad news associated with earnings, revenues, dividends, assets, credit ratings, stock prices, price targets, and analyst ratings. Bond price reaction is materialized by a strong negative movement on the day of the event (from $t=-1$ to $t=0$ with close-close-returns) and a continuation of the negative trend beyond the event day. Announcements associated with **cash flow generation (earnings, revenues, dividends)** generate an average underperformance of 15, 8, and 40 bps, respectively, in the two weeks following the bad news. This negative credit repricing indicates that the bad news was not incorporated in the bond prices, suggesting

that weaker-than-expected earnings announcements follow other bad news. Similarly, announcements associated with **credit quality (credit ratings, analyst ratings)** generate an average underperformance of 15 bps in the two weeks following the bad news. This suggests that credit risk agencies and analysts tend to downgrade a bond issuer following other bad news and negative past bond performance. When following good past bond performance, announcements generally lead to substantial underperformance on the day of the bad news, with nearly no post-event excess returns, particularly for events associated with **credit ratings, earnings, and equity actions**. Similarly, good news following negative past bond performance generally leads to substantial excess returns on the day of the news. This generates post-event outperformance, particularly for events associated with **credit and insider trading**. This suggests that unexpected announcements generate stronger bond price reactions in the short term. Interestingly, when following positive past bond performance, the impact of good news is generally negative on the day of the event, while excess cumulative average abnormal bond returns remain mostly flat after the event. This is in line with HUANG ET AL. (2020), who found that institutional bond investors tend to trade against positive news, in a “buy the rumor, sell the news” approach.

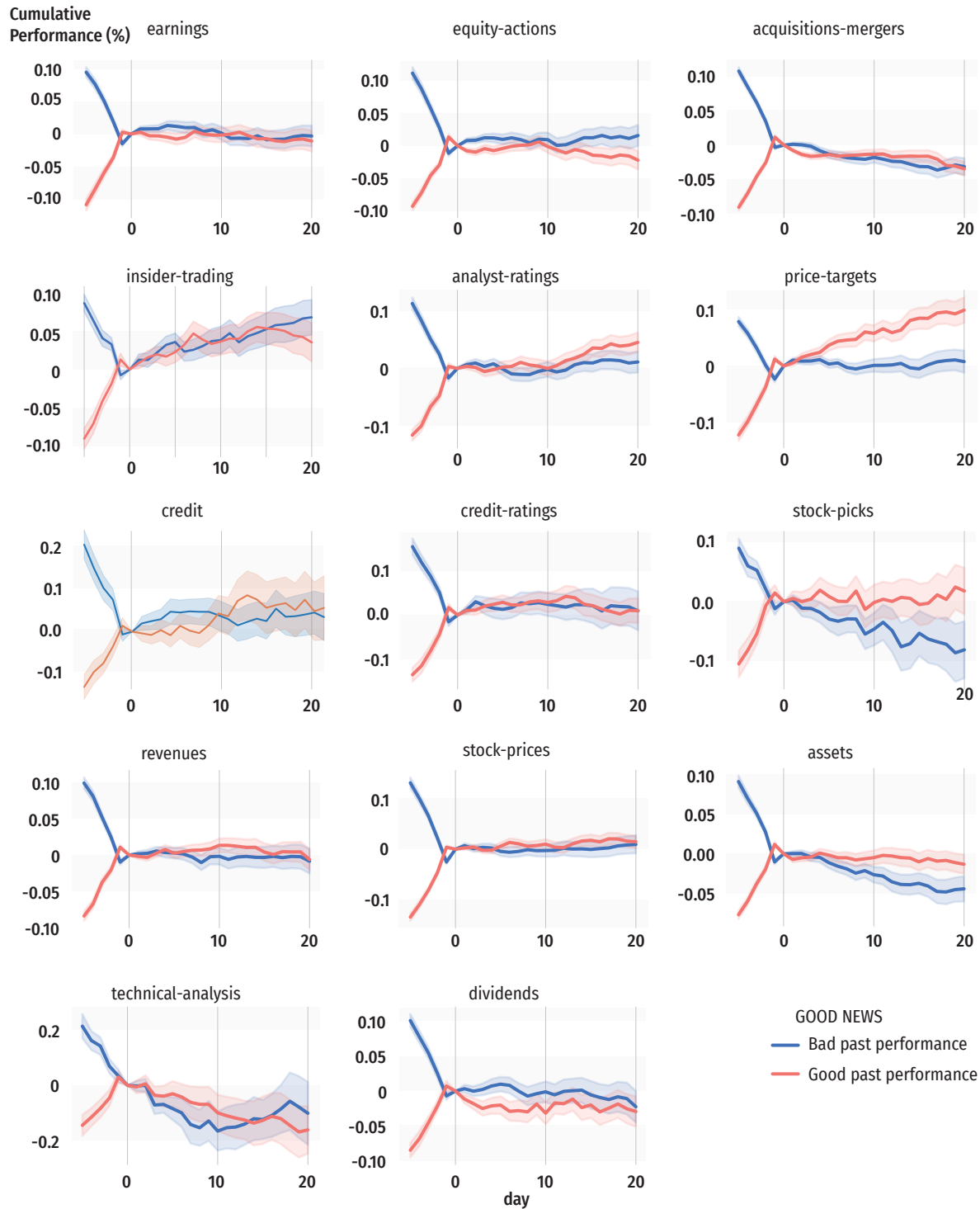


FIGURE 9A: Cumulative average abnormal returns, *CAAR*, throughout the event window for good news conditioned on good (red) and bad (blue) past performance, across groups of credit-related events. Shaded areas correspond to +/- 1 standard deviation confidence intervals. Bond returns are in excess of the bond benchmark of the same maturity and credit rating. Source: RavenPack, September 2022

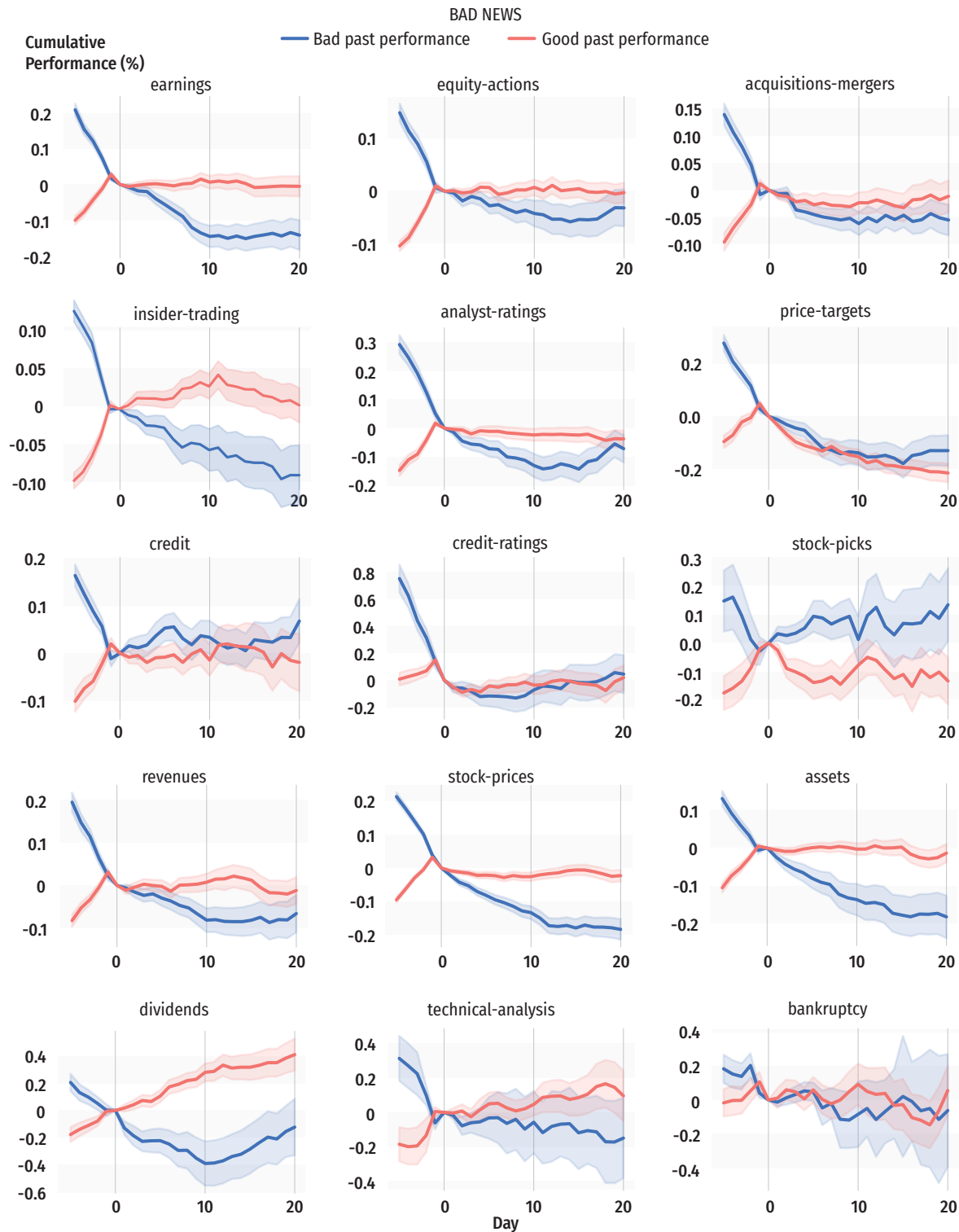


FIGURE 9B: Cumulative average abnormal returns, *CAAR*, throughout the event window for bad news conditioned on good (red) and bad (blue) past performance, across groups of credit-related events. Shaded areas correspond to ± 1 standard deviation confidence intervals. Bond returns are in excess of the bond benchmark of the same maturity and credit rating. Source: RavenPack, September 2022

5. Portfolio Construction

5.1 Benchmark Tilting

As individual credit instruments with more positive sentiment tend to outperform peers with less positive sentiment and vice versa (e.g. LOUGHRAN AND MCDONALD, 2016), the credit benchmark weights vector $W_t^L = [W_{1,T}^L, W_{2,T}^L, \dots, W_{N,T}^L]$ associated with N bonds at time t is tilted with a sentiment scores vector $s_{t+1} = [s_{1,t+1}, s_{2,t+1}, \dots, s_{N,t+1}]$. The tilted weight vector is calculated at the event category level on the same day, 30 minutes before the U.S. market close (3:30 p.m. EDT) to ensure that portfolio tilts are applied at the end of the trading session $t+1$. Credit sentiment scores are aggregated at the category level over smoothing periods of $\{1, 5, 10, 20\}$ days. The resulting tilted weights vector can be described as follows:

$$\tilde{W}_{j,t+1}^L = \delta_L \frac{W_{j,t}^L \exp(\omega s_{j,t+1})}{\sum_{i=1}^N W_{i,t}^L \exp(\omega s_{i,t+1})}$$

where ω is a constant scalar used as a scaling parameter to control for tilt magnitude ($\omega = 0$ is equivalent to no weighting tilt), $s_{j,t}$ is the aggregated sentiment score, and $\delta_L = 1$ ensures a gross portfolio exposure of 100%. The transformations above allow individual bonds with more extreme sentiment to have larger weights, since the exponential function tunes the tilting magnitude between normalized sentiment scores through different values of ω . We apply a higher max-

imum allocation constraint of 2.5% per bond instrument relative to the benchmark to allow for stronger tilting, using a recursive reallocation procedure whereby portfolio instruments that don't violate this constraint are repeatedly allocated additional weight proportional to the signal until the maximum gross exposure of 100% is achieved.

5.2 Standalone Performance Enhancement

In the remainder of the study, we remove earnings-related news (earnings, revenues, dividends) in line with HAFEZ ET AL. (2022), for simplicity. **FIGURE 10** shows the cumulative performance of the tilting-based credit strategies at the event category level when aggregating all news (top left), for sentiment scores **unconditioned by past performance or sentiment**. Tilting towards credit sentiment generally enhances excess returns relative to the U.S. investment-grade benchmark for larger values of tilting parameters ω .

Performance enhancement increases for strategies that aggregate news related to **analyst ratings, price targets, stock prices, and equity actions** over shorter time horizons, as price response preeminently occurs during post-announcement trades. Tilting towards events associated with **insider trading** and **credit** enhances excess performance to a lesser extent when aggregating news over longer periods, as we retain categories of events with the strongest bond price reaction, including insider sales, purchases, and debt sale, extension.

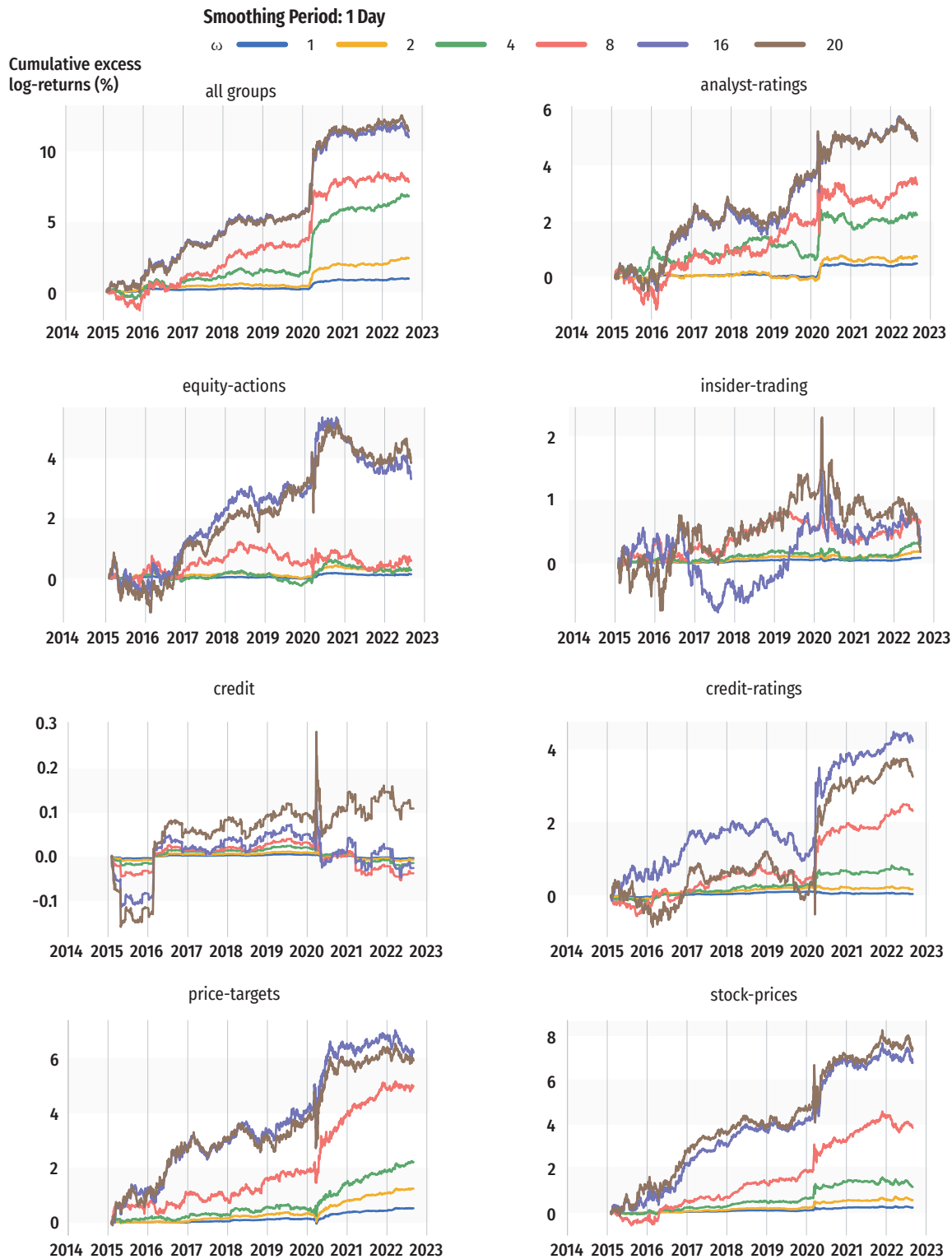


FIGURE 10A: Cumulative benchmark-adjusted performance comparisons of long-only credit portfolios tilted with credit sentiment scores for each category of credit-related events, for different tilting magnitudes, over different smoothing periods of $\{1\}$ days when rebalancing on a daily frequency across the U.S. investment-grade universe. Source: RavenPack, September 2022

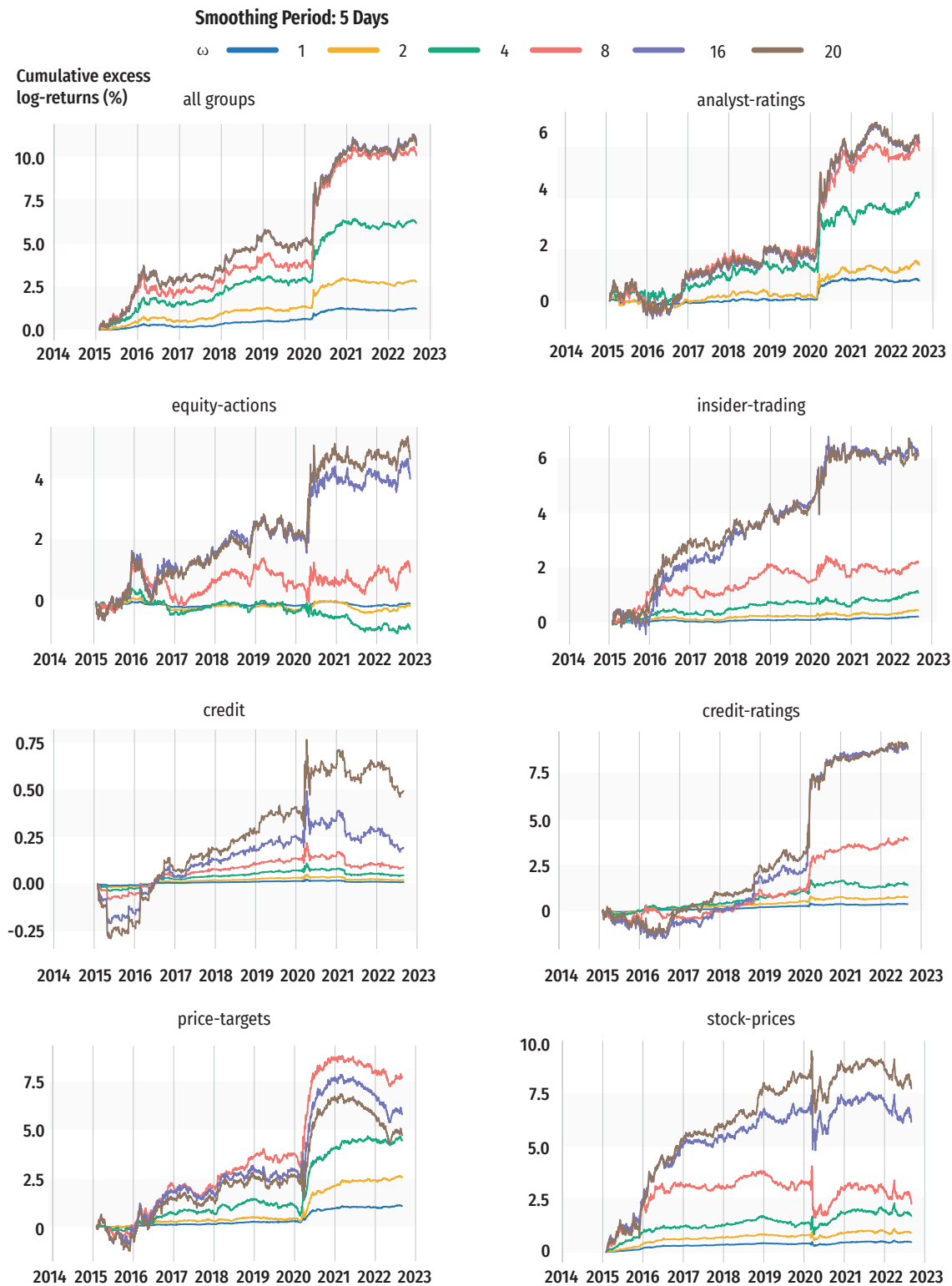


FIGURE 10B: Cumulative benchmark-adjusted performance comparisons of long-only credit portfolios tilted with credit sentiment scores for each category of credit-related events, for different tilting magnitudes, over different smoothing periods of $\{5\}$ days when rebalancing on a daily frequency across the U.S. investment-grade universe. Source: RavenPack, September 2022

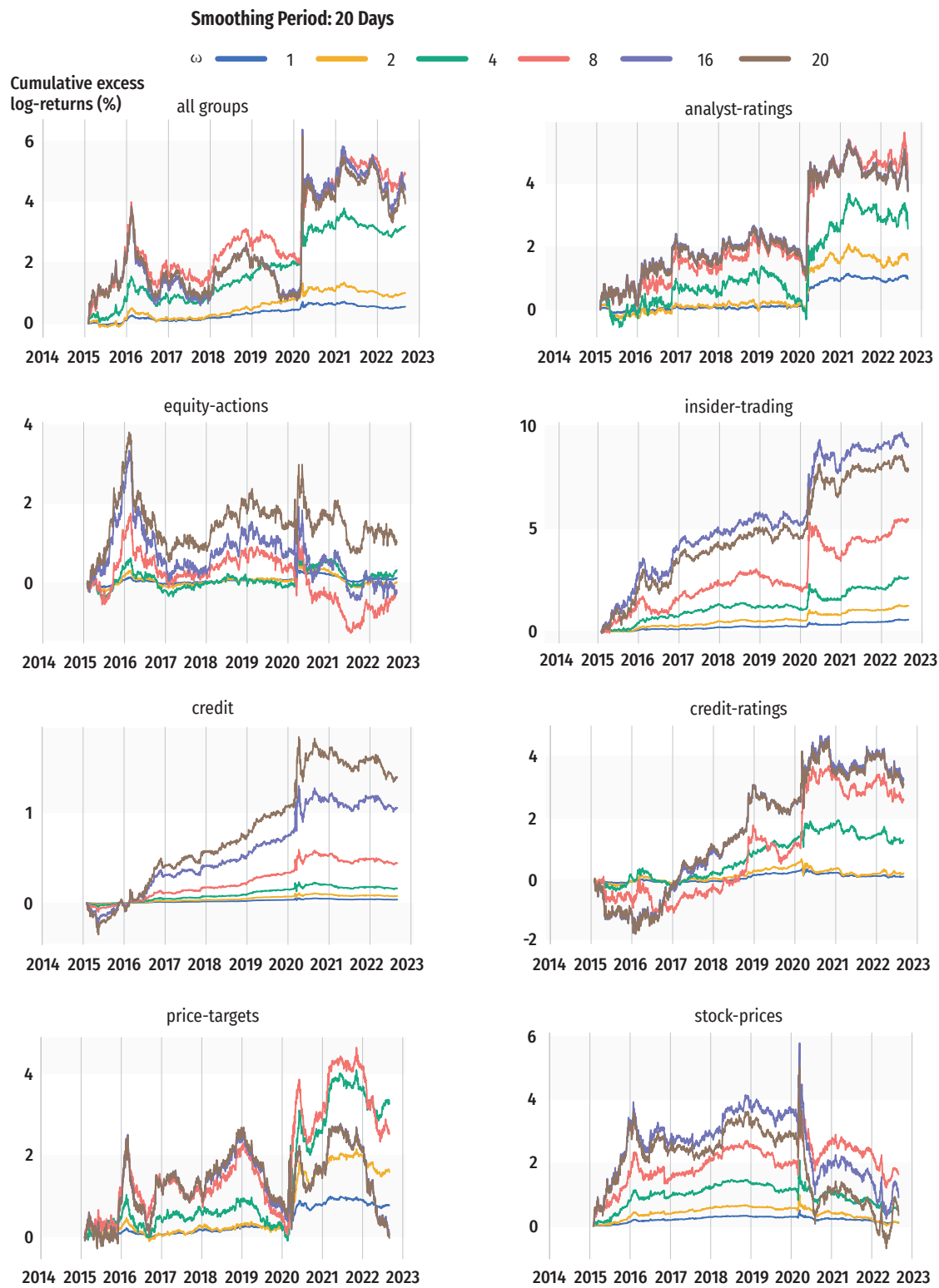


FIGURE 10C: Cumulative benchmark-adjusted performance comparisons of long-only credit portfolios tilted with credit sentiment scores for each category of credit-related events, for different tilting magnitudes, over different smoothing periods of {20} days when rebalancing on a daily frequency across the U.S. investment-grade universe. Source: RavenPack, September 2022

5.3 Conditional Performance Enhancement

In this section, we condition event selection based on past bond performance and news sentiment, as performed in the event studies. For comparison purposes, we estimate the effective holding period, EHP , in days, as the inverse function of portfolio turnover as defined by DEMIGUEL ET AL. (2009):

$$EHP = (T-1) \left(\sum_{t=1}^{T-1} \sum_{j=1}^N |W_{j,t+1} - W_{j,t}| \right)^{-1}$$

where $W_{i,t}$ denotes the weight associated to i -th security before rebalancing but at time $t+1$, and $W_{i,t+1}$ the portfolio weight at $t+1$ (after rebalancing). As observed in **FIGURE 11**, excess annualized returns are a decreasing (increasing) monotonic function of effective holding periods (portfolio turnover), showing the extent to which higher tilting magnitudes and shorter aggregation periods yield the maximum benefits from short-lived credit news.

FIGURE 11 shows the excess returns over different effective holding periods when conditioning separately on good versus poor past bond performance and good versus bad news at the event category level, over different aggregation periods and different tilting magnitudes (each dot). We see:

1. **An “upside surprise” effect: Conditioning credit events on good news and poor past performance generally outperforms the unconditional strategy over longer aggregation periods when tilting more aggressively.** At the monthly aggregating period (bottom left), tilting conditionally towards **credit ratings**, **equity actions**, and **stock prices**, respectively, generates excess annualized returns (AR) of 100 bps, 80 bps, and 30 bps compared to 40 bps, 20 bps, and 10 bps for unconditional tilting at the two-week effective holding period. **Analyst ratings** and **price targets** perform the best over the short term with returns of 80 bps in excess of the annual benchmark at the one-week

holding period. The unconditional tilting strategy performs the best at the two-week effective holding period with excess AR of nearly 40 bps.

2. **A “downside surprise” effect: Conditioning credit ratings, analyst ratings, and price targets on bad news and good past performance generates long-lasting outperformance.** Tilting towards **price targets** and **analyst ratings** at the monthly aggregating period generates annualized returns of 18 bps and 10 bps in excess of the annual credit benchmark respectively, at the two-month effective holding period. Tilting towards **credit downgrades** at the daily aggregating period generates an extra return of 5 bps at the seven-month effective holding period, evidencing a long-lasting credit repricing.
3. **A “buy the rumor, sell the news” effect: Conditioning credit events on good news and good past performance always underperforms the unconditional strategy over longer aggregation periods.** At the monthly aggregating period, tilting conditionally towards **credit ratings**, **equity actions**, **analyst ratings**, **price targets**, and **stock prices** generates negative excess AR, underperforming the unconditional tilting strategy by 50 bps, 60 bps, 40 bps, 35 bps, and 40 bps, respectively, at the two-week effective holding period. Tilting towards **credit** and **insider trading** underperforms the unconditional tilting strategy by nearly 20 bps at the two-month effective period.

In **FIGURE 12**, the “buy the rumor, sell the news” strategy significantly underperforms both the credit benchmark and the unconditionally tilted strategy at all aggregating periods and effective holding periods. At the one-week effective holding period, when aggregating over 10 days, it underperforms by 20 bps and 100 bps, respectively. However, the combined “upside surprise” strategy performs well over short-term horizons when aggregating over longer periods. At the monthly aggregating period, the combined strategy outperforms the unconditionally tilted strategy by nearly 80 bps at the one-week holding period.

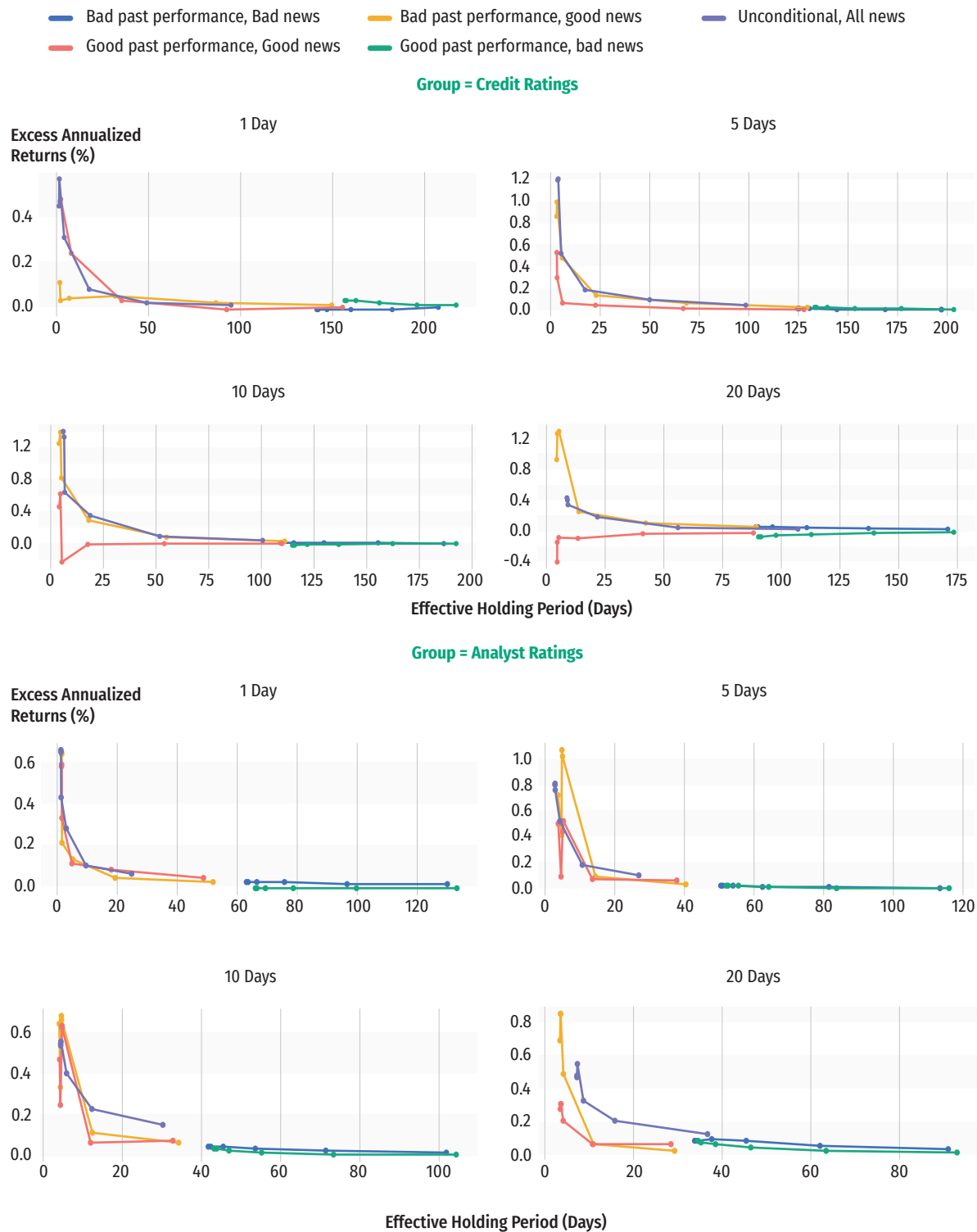


FIGURE 11A: Excess annualized returns for various effective holding periods at the events category level for long-only credit portfolios tilted with credit sentiment scores over different tilting magnitudes ω , over different smoothing periods of $\{1, 5, 10, 20\}$ days, when rebalancing on a daily frequency across the U.S. investment-grade universe. Source: RavenPack, September 2022

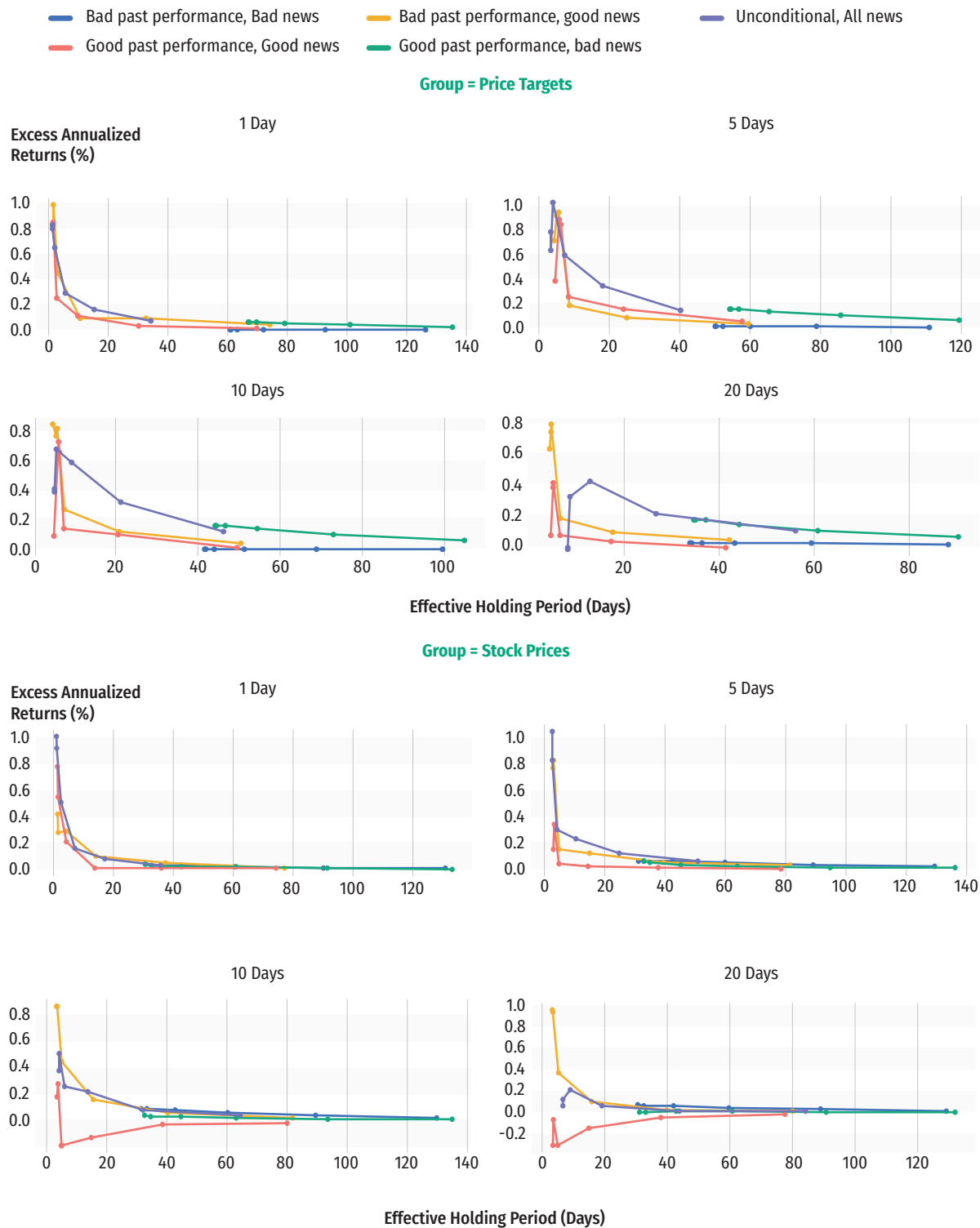


FIGURE 11B: Excess annualized returns for various effective holding periods at the events category level for long-only credit portfolios tilted with credit sentiment scores over different tilting magnitudes ω , over different smoothing periods of $\{1, 5, 10, 20\}$ days, when rebalancing on a daily frequency across the U.S. investment-grade universe. Source: RavenPack, September 2022

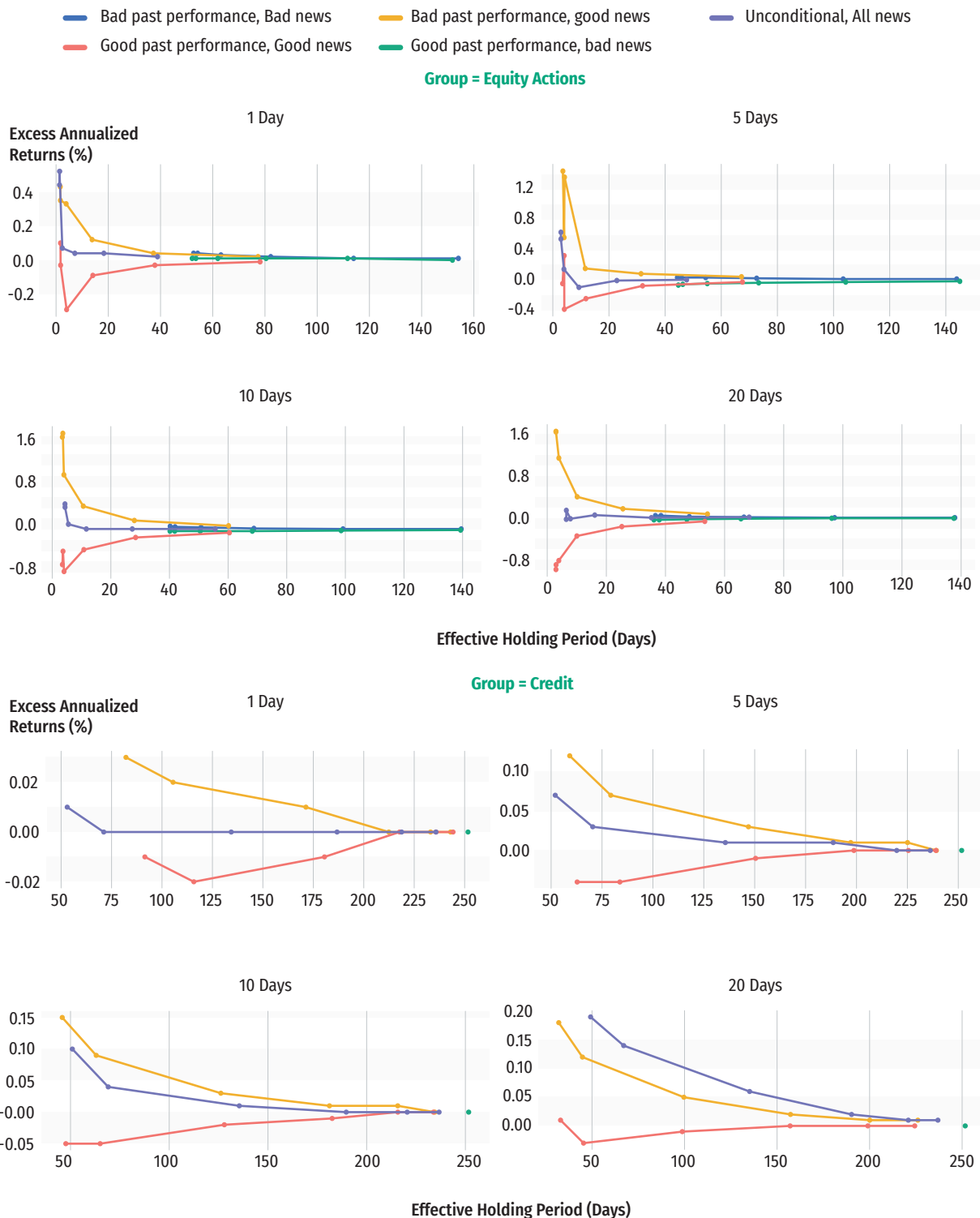


FIGURE 11C: Excess annualized returns for various effective holding periods at the events category level for long-only credit portfolios tilted with credit sentiment scores over different tilting magnitudes ω , over different smoothing periods of $\{1, 5, 10, 20\}$ days, when rebalancing on a daily frequency across the U.S. investment-grade universe. Source: RavenPack, September 2022

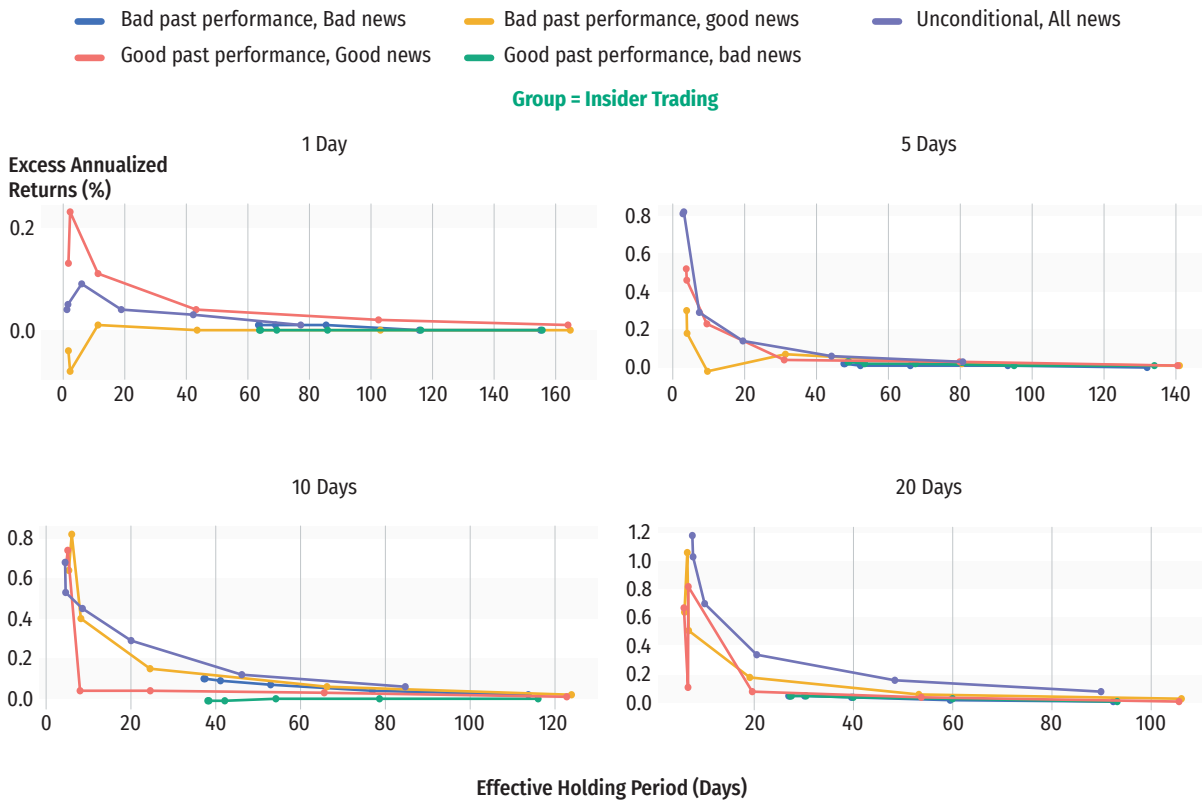


FIGURE 11D: Excess annualized returns for various effective holding periods at the events category level for long-only credit portfolios tilted with credit sentiment scores over different tilting magnitudes ω , over different smoothing periods of $\{1, 5, 10, 20\}$ days, when rebalancing on a daily frequency across the U.S. investment-grade universe. Source: RavenPack, September 2022

As observed in **FIGURE 13**, the **combined strategy**, which excludes good announcements associated with good past performance, while targeting unexpected events, always outperforms the unconditional strategy. Aggregating credit sentiment scores over longer periods enhances the short-term performance of the strategy. At the monthly aggregating period, it generates annualized returns of 200 bps in excess of the annual credit benchmark at a two-day effective holding period. At the one-week holding period, the combined strategy generates excess AR of 90 bps compared to 50 bps for the unconditional one.

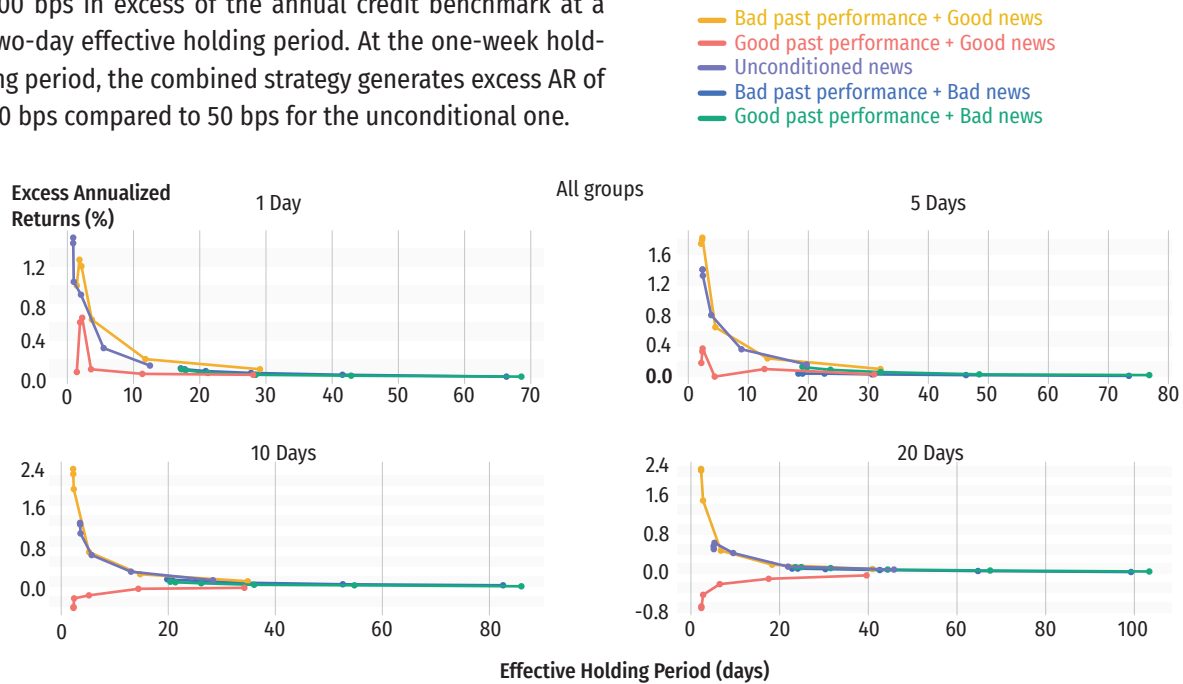


FIGURE 12: Excess annualized returns over various effective holding periods for long-only credit portfolios tilted with credit sentiment scores aggregated at the bond issuer level over different tilting magnitudes ω , over different smoothing periods of $\{1, 5, 10, 20\}$ days, when rebalancing on a daily frequency across the U.S. investment-grade universe. Source: RavenPack, September 2022

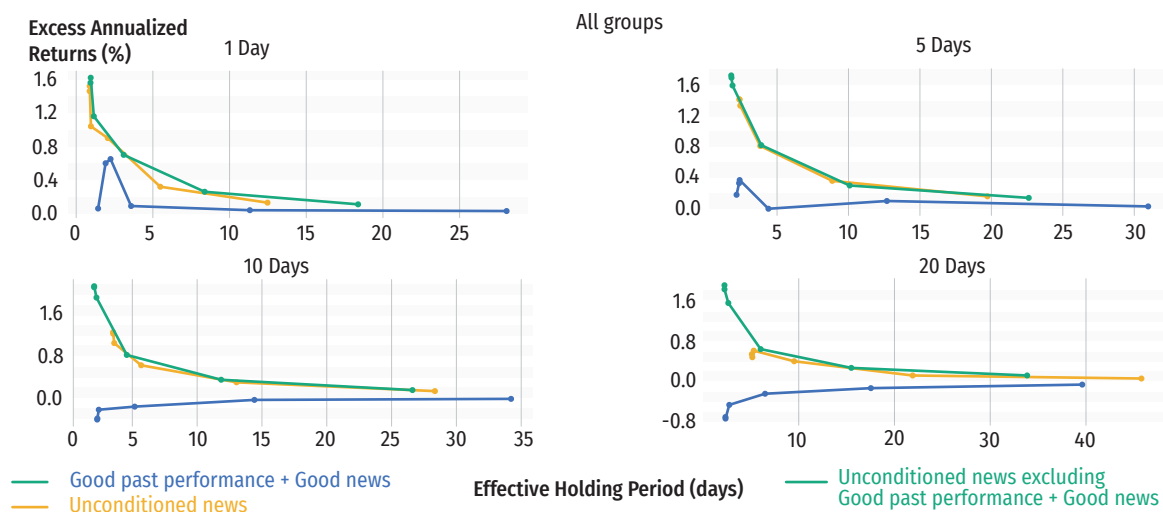


FIGURE 13: Excess annualized returns of the combined strategy (green) over various effective holding periods for long-only credit portfolios tilted with credit sentiment scores aggregated at the bond issuer level. Source: RavenPack, September 2022

6. Conclusion

In this study, we performed event studies to evaluate the impact of granular credit-related news on fixed income prices. We demonstrated that credit-related news has a significant market impact on bond prices, particularly when conditioning on news sentiment and past performance. Conditioning announcements associated with credit repricing (credit ratings, analyst ratings, and price targets) on bad news and good past performance generated long-lasting outperformance, with news incorporated slowly into bond prices. Good news following negative past bond performance generally led to substantial abnormal returns on the day of the news while also generating post-event outperformance, particularly for events associated with credit and insider trading. This suggests that unexpected announcements generate stronger bond price reactions in the short term. Although we observed that credit risk agencies and analysts tend to downgrade

bond issuers following bad news and negative past bond performance, suggesting an element of negative news momentum, we found that bond investors tend to trade against positive news, in a “buy the rumor, sell the news” approach. Leveraging these results, the **combined strategy**, which excluded good announcements associated with good past performance while targeting unexpected events, always outperformed the unconditional strategy. Aggregating credit sentiment scores over longer periods enhanced the short-term performance of the combined strategy. At the monthly aggregating period, it generated annualized returns of 200 bps in excess of the annual credit benchmark at a two-day effective holding period. At the one-week holding period, the combined strategy generated excess AR of 90 bps compared to 50 bps for the unconditional strategy.

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